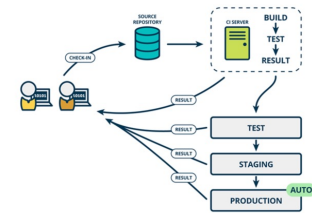


Continuous Integration/Continuous Development

COMS/SE 4170
Spring 2026

1

CI/CD



The CI/CD Process

<https://www.mindtheproduct.com/what-the-hell-are-ci-cd-and-devops-a-cheatsheet-for-the-rest-of-us/>

2

Continuous Integration (CI/CD) in gitlab

Create a .gitlab-ci.yml file

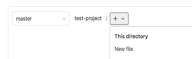
Now create a .gitlab-ci.yml file. It is a YAML file where you specify instructions for GitLab CI/CD.

In this file, you define:

- The structure and order of jobs that the runner should execute.
- The decisions the runner should make when specific conditions are encountered.

To create a .gitlab-ci.yml file:

1. On the left sidebar, select **Code > Repository**.
2. Above the file list, select the branch you want to commit to. If you're not sure, leave **default** or **main**. Then select the plus icon (+) and **New file**.

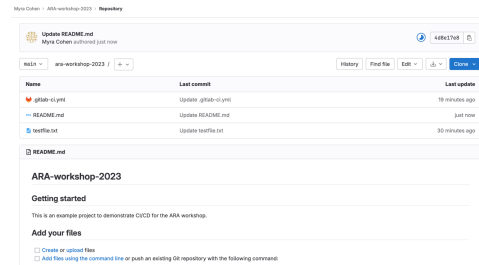


Nice resource for starting with CI/CD in gitlab:

- https://docs.gitlab.com/ee/ci/quick_start/index.html

3

Example (gitlab project)



4

Example (set up CI/CD)

```

1 update_github-ci.yml
2 stage: build
3 script:
4   - echo "hello, BITLAB_USER_LOGIN!"
5
6
7 test-job1
8 stage: test
9 script:
10  - echo "This job demonstrates a test running in CI-CD for the AAA-workshop 2023!"
11
12 test-job2
13 stage: test
14 script:
15  - echo "This job tests something else, but takes more time than test-job1."
16  - echo "After the echo commands complete, it runs the sleep command for 20 seconds"
17  - echo "which simulates a test that runs 20 seconds longer than test-job1"
18  - sleep 20
19
20 deploy-prod
21 stage: deploy
22 script:
23  - echo "This job deploys something from the BCL_COMMIT_BRANCH branch."
24  - echo "production"
25
26

```

5

Example

fe8f279de8b3...

You can move around the graph

Git revision

Showing 1 changed file - with 2 additions and 2 deletions

Changes Pipelines

Showing 1 changed file - with 2 additions and 2 deletions

Hide with

Preview

6

Example

Myra Cohen · AAA-workshop-2023 · Pipelines

All Finished Branches Tags

Project overview

Filter pipelines

Status	Pipeline	Triggerer	Stages
passed	Update github-ci.yml	#33005 · f88f279e	build test
passed	Update terraform	#33004 · f88f279e	build

Pipelines

- Jobs
- Pipeline editor
- Pipeline schedules
- Artifacts

7

Example

Myra Cohen · AAA-workshop-2023 · Pipelines

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Pipelines

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8

Continuous Integration (CI/CD) in gitlab

```

1 Running with gitlab-runner 16.1.0 (86d94642)
2 on ara-workshop2023 hywxy05, system ID: s_Mha3d8dc09
3 Preparing the "shell" executor...
4 Using Shell (bash) executor...
5 Preparing environment
6 Running on local...
7 Setting source from Git repository
8 Fetching changes with git depth set to 50
9 Initializing existing Git repository in /builds/hywxy05/ara-workshop-2023/.git/
10 Checking out f0b7279a as detached HEAD (ref is main)...
11 Checking out main branch
12 Executing "test_script" stage of the job script
13 $ echo "This job demonstrates a test running in CI-CD for the ARA-Workshop 2023"
14 This job demonstrates a test running in CI-CD for the ARA-Workshop 2023
15 Cleaning up project directory and file based variables
16 Job succeeded
  
```

9

Continuous Integration (CI/CD) in gitlab

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1 Running with gitlab-runner 16.1.0 (86d94642)
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14 This job demonstrates a test running in CI-CD for the ARA-Workshop 2023
15 Cleaning up project directory and file based variables
16 Job succeeded
  
```

10

Integration Failures

Project: Myra Cohen / ARA-v

Branch: ☐ main

Commit: ☐ e9f0880a Update gitlab-ci.yml

Commit Author: Myra Cohen

Pipeline #33089 triggered by test

Pipeline #33089 triggered by Myra Cohen successfully completed 5 jobs in 3 stages.

Failed job: test

11

Example C++ (.gitlab-ci.yml) template

```

14 image: gcc
15
16 build:
17   # Instead of calling g++ directly you can also use some build toolkits like make
18   # Install the necessary build tools when needed
19   # before_script:
20   #   - apt update && apt -y install make autoconf
21   script:
22     - g++ -std=c++11 -c main.cpp -o mybinary
23   artifacts:
24     paths:
25       - mybinary
26     # depending on your build setup it's most likely a good idea to cache outputs to reduce the build time
27     # cache:
28     #   paths:
29     #     - "*/a*"
30
31 # run tests using the binary built before
32 test:
33   stage: test
34   script:
35     - ./runmytests.sh
36
37 deploy:
38   stage: deploy
39   script: echo "Define your deployment script!"
40   environment: production
  
```

12

Adding a C++ Test to our Project

```

10 test-job2:
11   stage: test
12   script:
13     - echo "This job tests s
14     - echo "After the echo c
15     - echo "Which simulates
16     - sleep 5
17
18 test-job3:
19   stage: test
20   script:
21     - chmod +x test1.sh
22     - ./test1.sh
23
24 deploy-prod:

```

Script in repository

yaml file

13

Adding a C++ Test to our Project

Name	Last commit
.gitlab-ci.yml	Update .gitlab-ci.yml
README.md	Update README.md
C++ helloARA.cpp	Update helloARA.cpp
test1.sh	Update test1.sh
testfile.txt	Update testfile.txt

14

Breaking the Build

```

C++ helloARA.cpp 101 B
1 #include <iostream>
2
3 int main() {
4   std::cout << "Hello ARA Workshop Participants!;
5   return 0;
6 }
7

```

Missing quote

15

Result

Status Pipeline T

failed failed Myra Cohen triggered pipeline for commit 9d11544 finished 3 minutes ago

For nix

60.5 Jobs 16 seconds, queued for 3 seconds

Pipeline Needs Jobs Failed Jobs Tests

build test deploy

build-job test-job1 test-job2 test-job3 deploy-prod

16

